



Analytic Continuation & SmoQyDEAC.jl

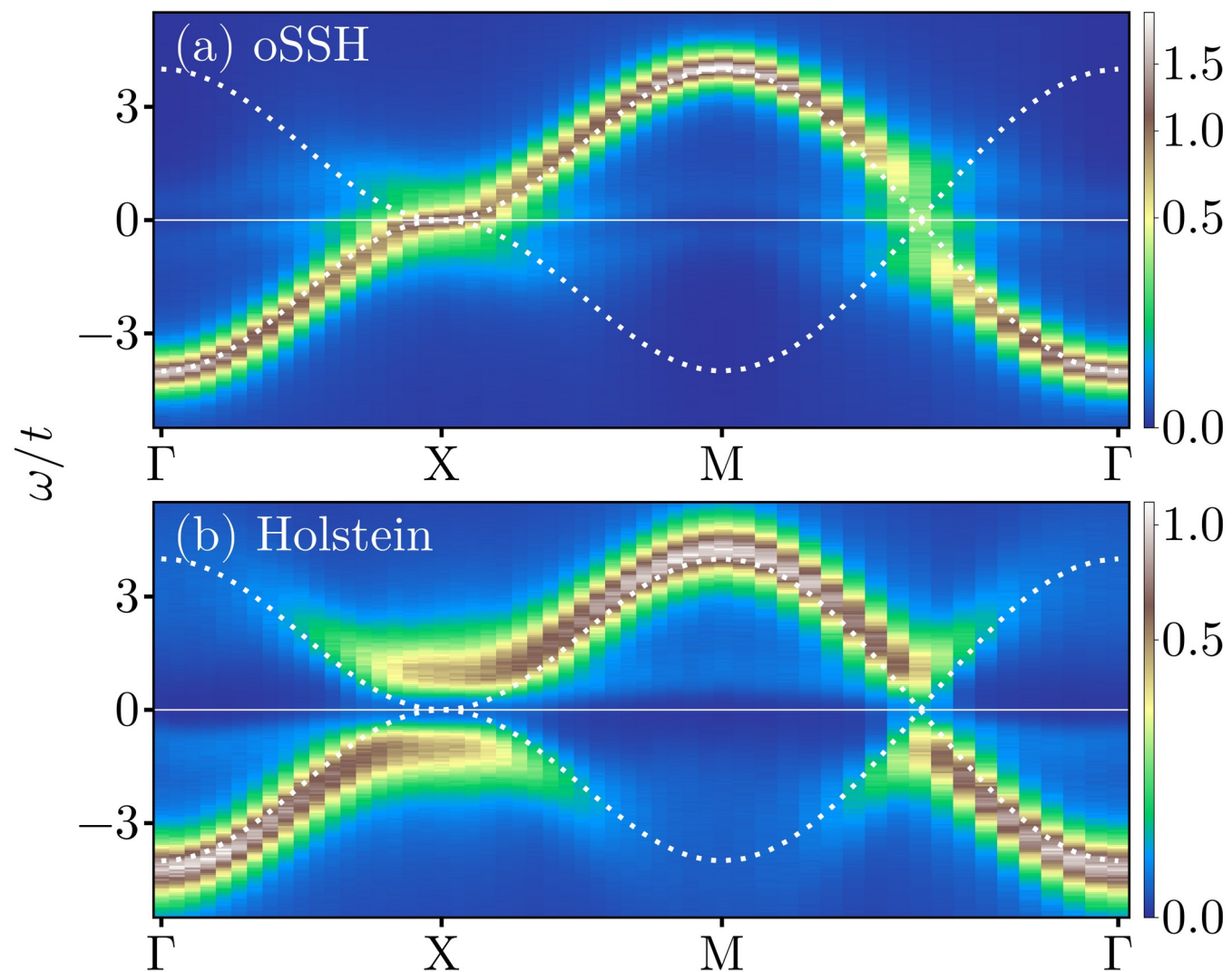
UC Davis



About me:



Amuse Bouche



Agenda

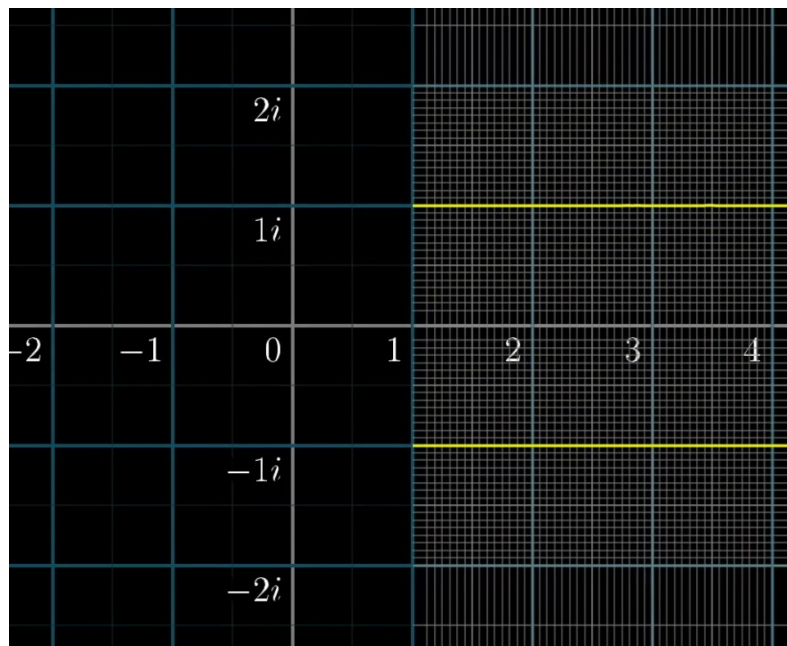
- What is Analytic Continuation
- MaxEnt Primer
- Other Methods
- DEAC algorithm
- Using SmoQyDEAC.jl

1. What is Analytic Continuation?

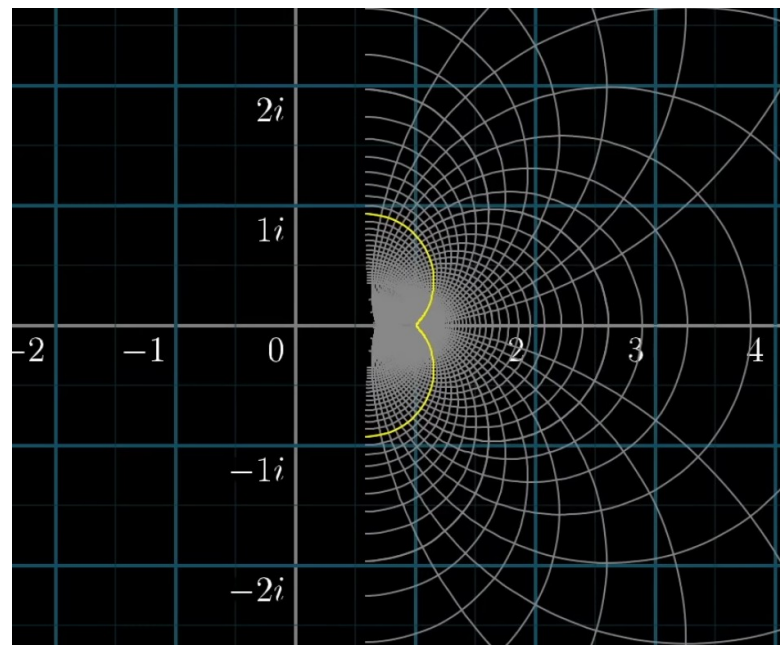
- Extends complex analytic function from one domain to larger domain
- An analytic function can be written as $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ for some neighborhood $r > 0, |z - z_0| < r$
- For complex functions this means they are differentiable everywhere they are defined
- New mapped function is meromorphic and **UNIQUELY DEFINED***
 - Meromorphic: analytic everywhere except may contain poles

Euler Zeta Function

$$\Re(z) > 1$$



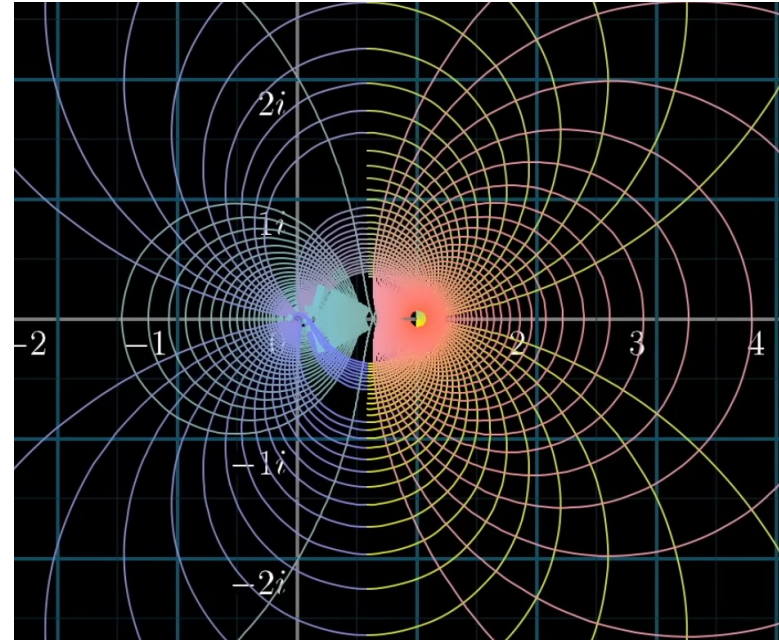
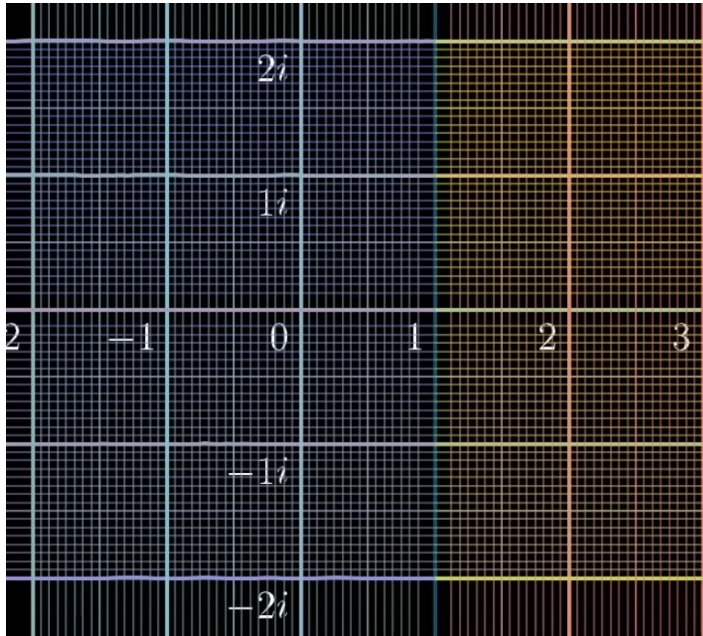
$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$



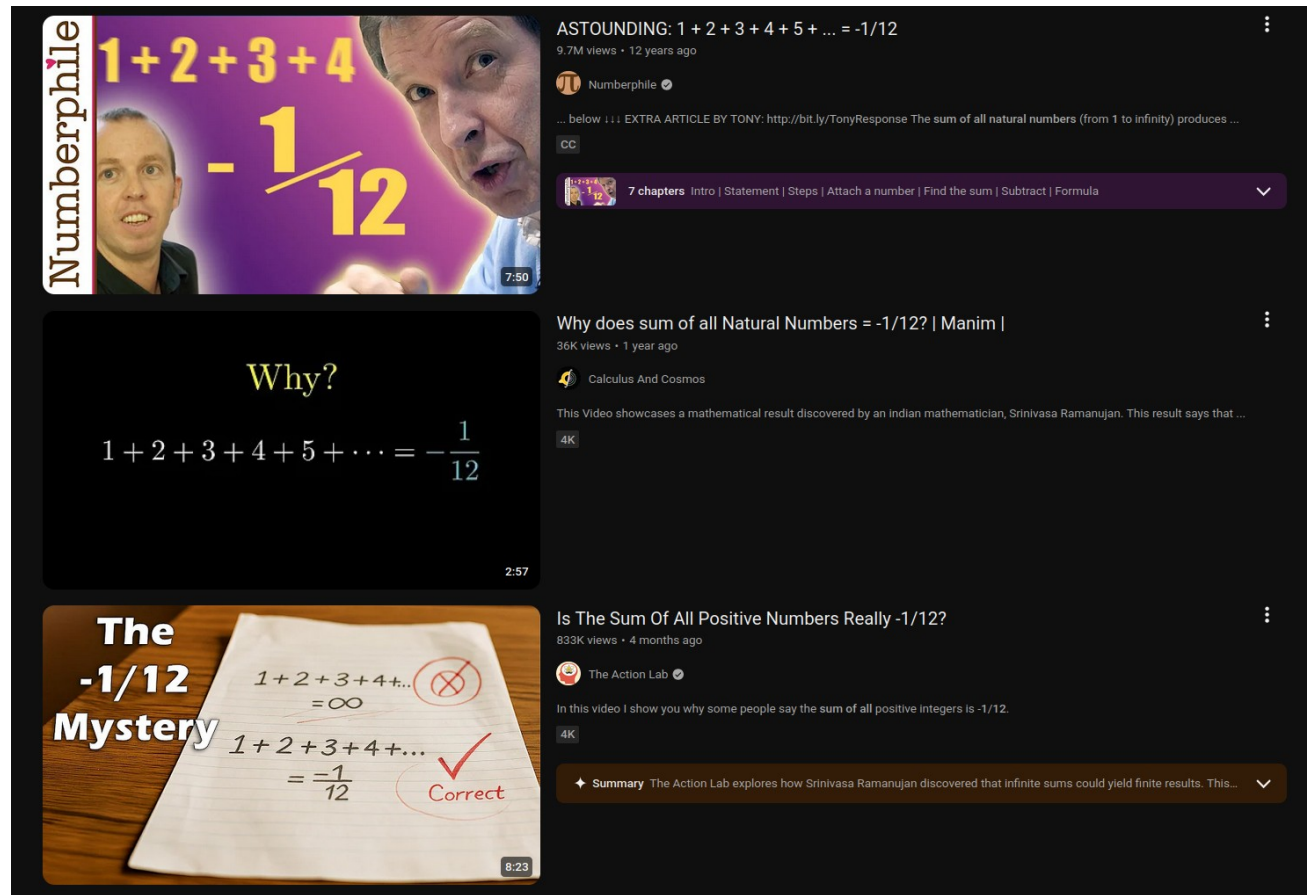
Riemann Zeta Function

z

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$



YouTube “Math”



The screenshot shows a YouTube search results page for the Ramanujan summation. The top result is from the channel 'Numberphile', titled 'ASTOUNDING: $1 + 2 + 3 + 4 + 5 + \dots = -1/12$ ', with 9.7M views and posted 12 years ago. The video thumbnail features the Numberphile logo, the equation $1+2+3+4 - 1/12$, and a man's face. Below the video player, there is a section titled 'Why?' showing the equation $1 + 2 + 3 + 4 + 5 + \dots = -\frac{1}{12}$. The second result is from 'Calculus And Cosmos', titled 'Why does sum of all Natural Numbers = -1/12? | Manim |', with 36K views and posted 1 year ago. The third result is from 'The Action Lab', titled 'Is The Sum Of All Positive Numbers Really -1/12?', with 833K views and posted 4 months ago. The video thumbnail for this result shows a piece of paper with the equation $1 + 2 + 3 + 4 + \dots = \infty$ crossed out and $1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$ marked as 'Correct' with a red checkmark.

Numberphile

ASTOUNDING: $1 + 2 + 3 + 4 + 5 + \dots = -1/12$

9.7M views • 12 years ago

Numberphile

... below : EXTRA ARTICLE BY TONY: <http://bit.ly/TonyResponse> The sum of all natural numbers (from 1 to infinity) produces ...

CC

7 chapters Intro | Statement | Steps | Attach a number | Find the sum | Subtract | Formula

Why?

$1 + 2 + 3 + 4 + 5 + \dots = -\frac{1}{12}$

2:57

The

-1/12

Mystery

$1 + 2 + 3 + 4 + \dots = \infty$

$1 + 2 + 3 + 4 + \dots = -\frac{1}{12}$

Correct

8:23

Why does sum of all Natural Numbers = -1/12? | Manim |

36K views • 1 year ago

Calculus And Cosmos

This Video showcases a mathematical result discovered by an Indian mathematician, Srinivasa Ramanujan. This result says that ...

4K

Is The Sum Of All Positive Numbers Really -1/12?

833K views • 4 months ago

The Action Lab

In this video I show you why some people say the sum of all positive integers is -1/12.

4K

Summary The Action Lab explores how Srinivasa Ramanujan discovered that infinite sums could yield finite results. This...

We aren't so lucky

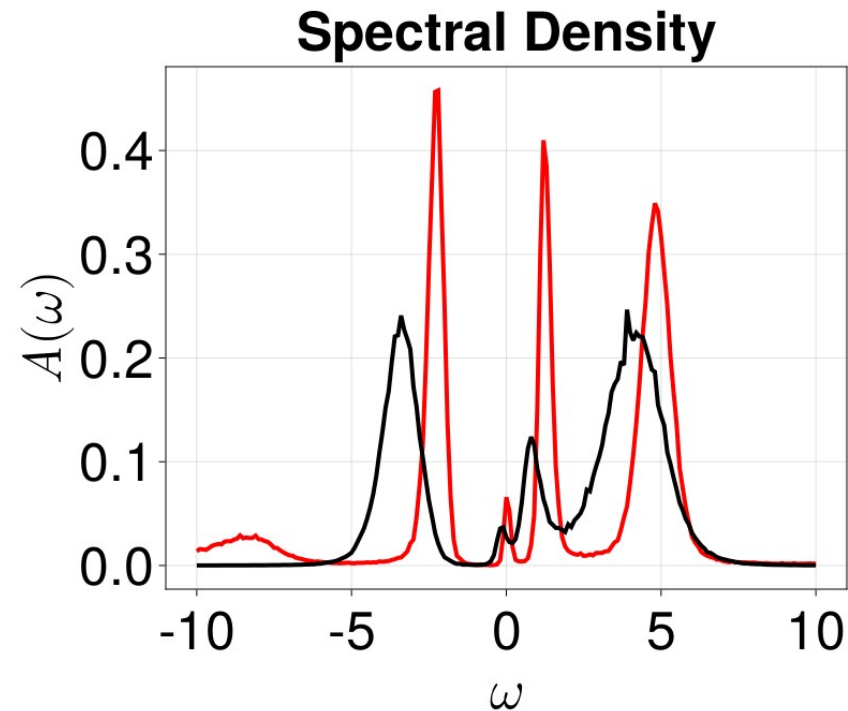
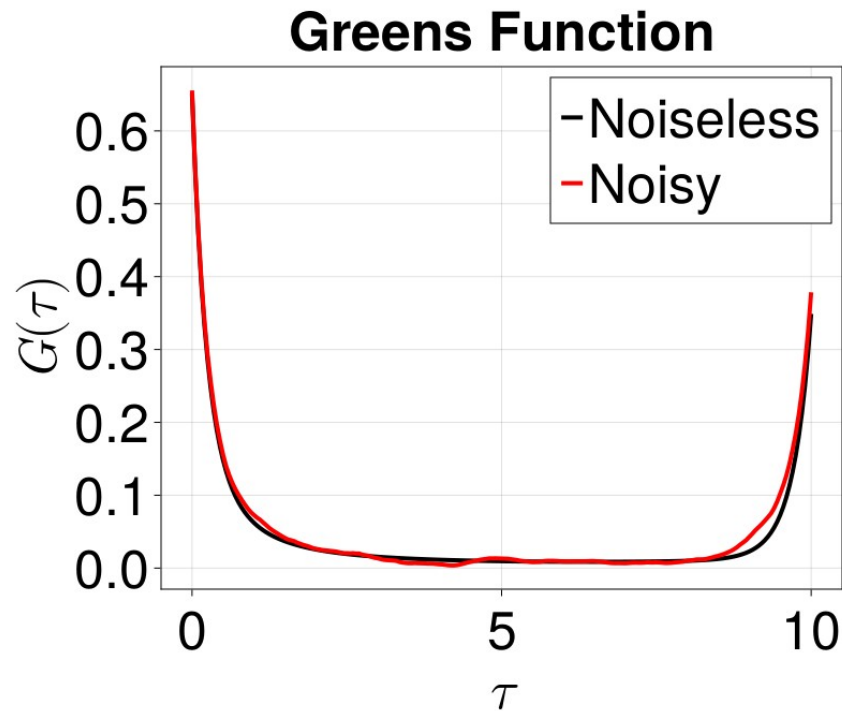
- Retardation means Green's functions are analytic on the upper half of the complex plane, but...
- QMC only takes measurements at discrete points on imaginary axis!
- QMC is also noisy
- Using just MC output we can no longer ensure uniqueness
- Naive attempts at AC will give garbage results

AC on QMC

$$G(i\omega_n) = \int_{-\infty}^{\infty} d\omega \frac{1}{i\omega_n - \omega} A(\omega)$$

$$G(\tau) = \int_{-\infty}^{\infty} d\omega \frac{e^{-\tau\omega}}{1 \pm e^{-\beta\omega}} A(\omega)$$

$$\mathbf{G} = \mathbf{K}\mathbf{A}$$



2. MaxEnt

M. Jarrell and J. Gubernatis, Bayesian inference and the analytic continuation of imaginary-time quantum Monte Carlo data, *Physics Reports* **269**, 133 (1996).

- Utilizes “model” and entropy term to fight overfitting to noise
- Given α , maximize posterior likelihood Q

$$Q = \alpha S - \frac{\chi^2}{2}$$

Entropy

$$S = \int d\omega \left(A(\omega) - M(\omega) - A(\omega) \log \left[\frac{A(\omega)}{M(\omega)} \right] \right)$$

$$S = \sum_i \Delta\omega \left(A(\omega_i) - M(\omega_i) - A(\omega_i) \log \left[\frac{A(\omega_i)}{M(\omega_i)} \right] \right)$$

Fitness

$$\chi^2 = \sum_{i,j}^L (\bar{G}_i - K_{ik} A_k) [C^{-1}]_{ij} (\bar{G}_j - K_{jk} A_k)$$

$$\chi^2 \approx \sum_i^L \frac{(\bar{G}_i - K_{ik} A_k)^2}{\sigma_i^2}$$

2. MaxEnt (cont)

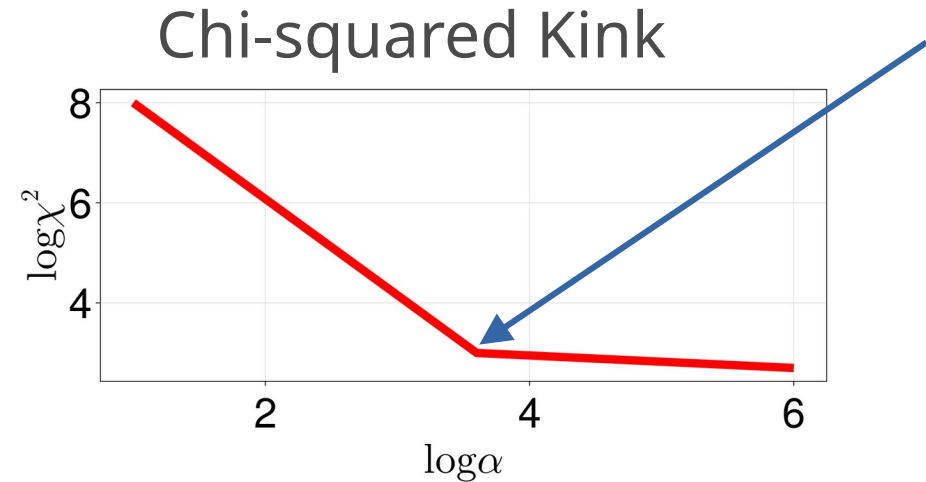
- User inputs array of alphas and default model M
- For each alpha MaxEnt uses solver to find A which maximizes Q
- Based on results for each alpha value, A is determined (next slide)

MaxEnt (cont)

- 2 Free parameters, $M(\omega)$ and α
- Result is typically not terribly reliant on choice of $M(\omega)$ (YMMV)
 - I usually just use a wide Gaussian centered on 0
- Alpha needs to be found

Bryan Algorithm

$$\bar{A}(\omega) = \int d\alpha A_{\alpha}(\omega) P[\alpha | \bar{G}]$$



ana_cont

- My go-to MaxEnt code
- Python based
 - I hate Python, so I run it in Julia using PyCall without issue
- Can utilize full covariance matrix!
- Example Python and Julia scripts in GitHub repo

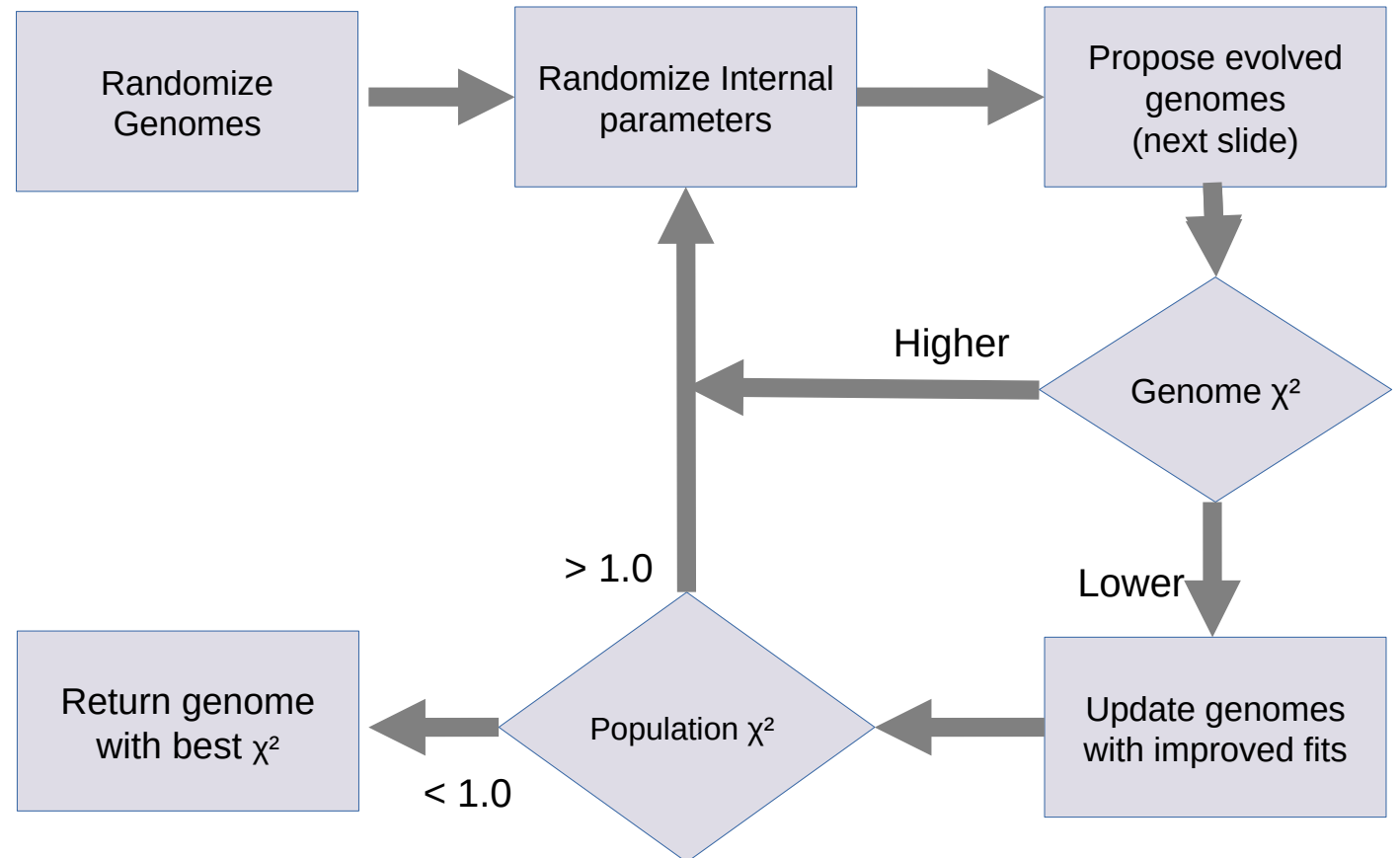
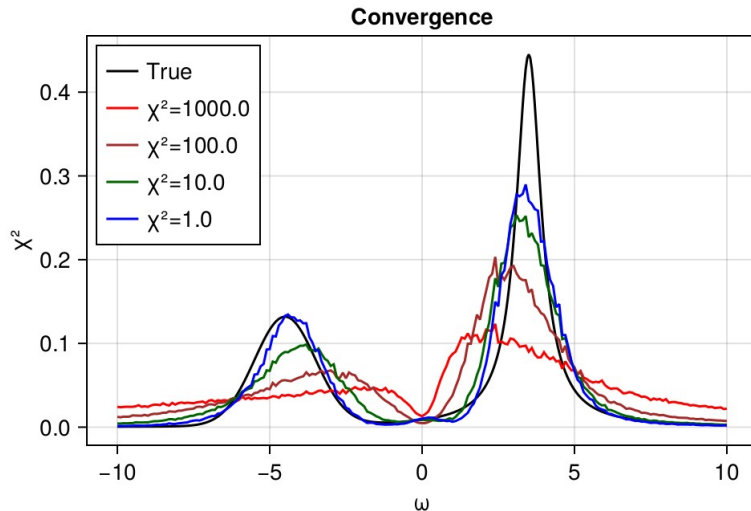
3. The AC Method Menagerie

- Nevanlinna – Not Noise resistant
- StochasticAC – Sandvik group: publishes papers on method but not code
- StochasticOM – I haven't found a good, usable code
- PronyAC – Emanuel Gull's group, not noise resistant
- Padé Approximation – Not Noise resistant
- Machine Learning methods – **Nobody** I trust to solve the problem is even working on this

4. Differential Evolution for AC

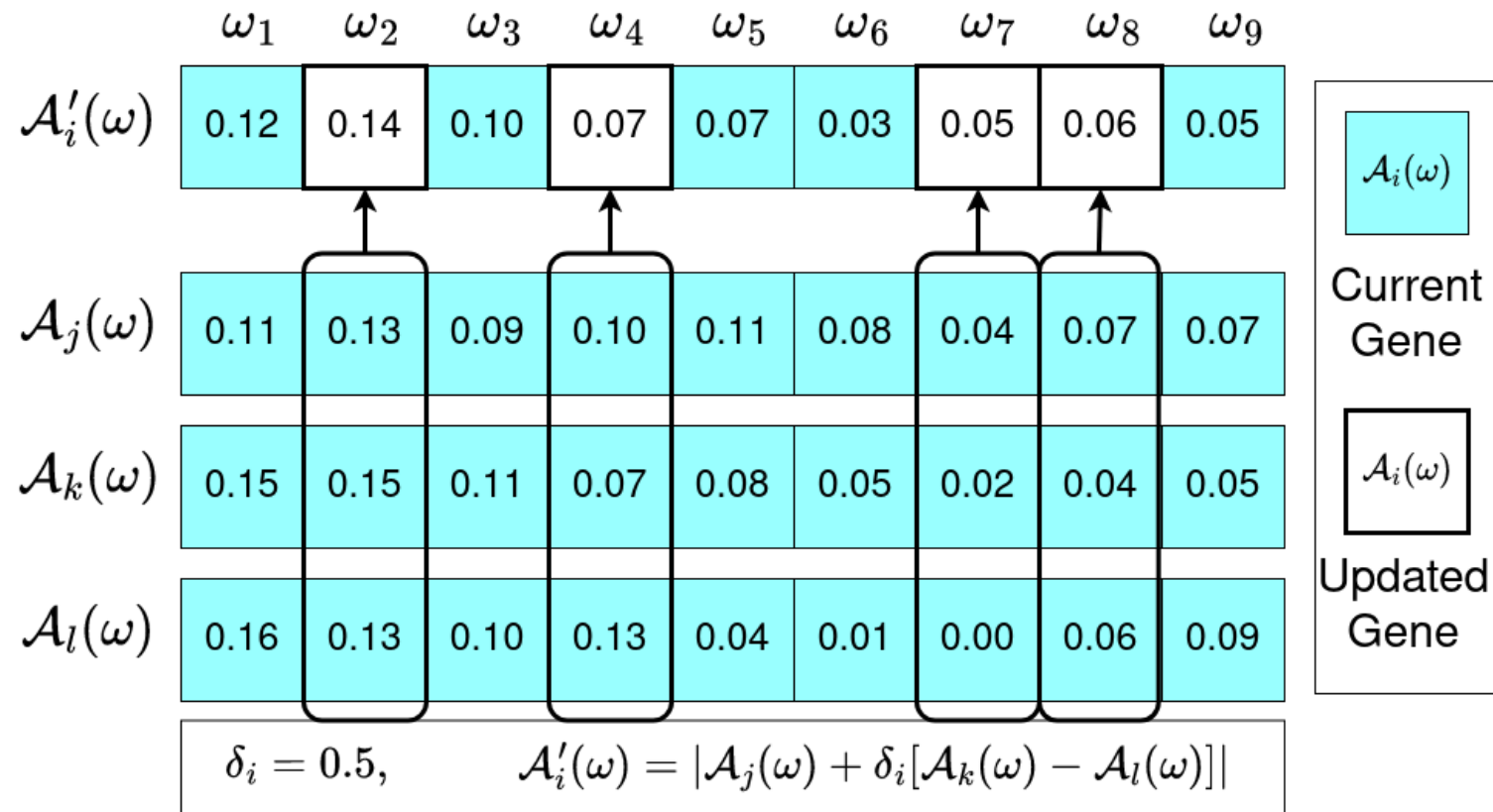
$$G_{\text{Calc}}(\tau_\alpha) = K(\tau_\alpha, \omega) A(\omega)$$

$$\chi^2 = \frac{1}{N} \sum_{\alpha=1}^{N=\beta/\Delta\tau} \frac{(G_{\text{Meas}}(\tau_\alpha) - G_{\text{Calc}}(\tau_\alpha))^2}{\sigma_i^2}$$



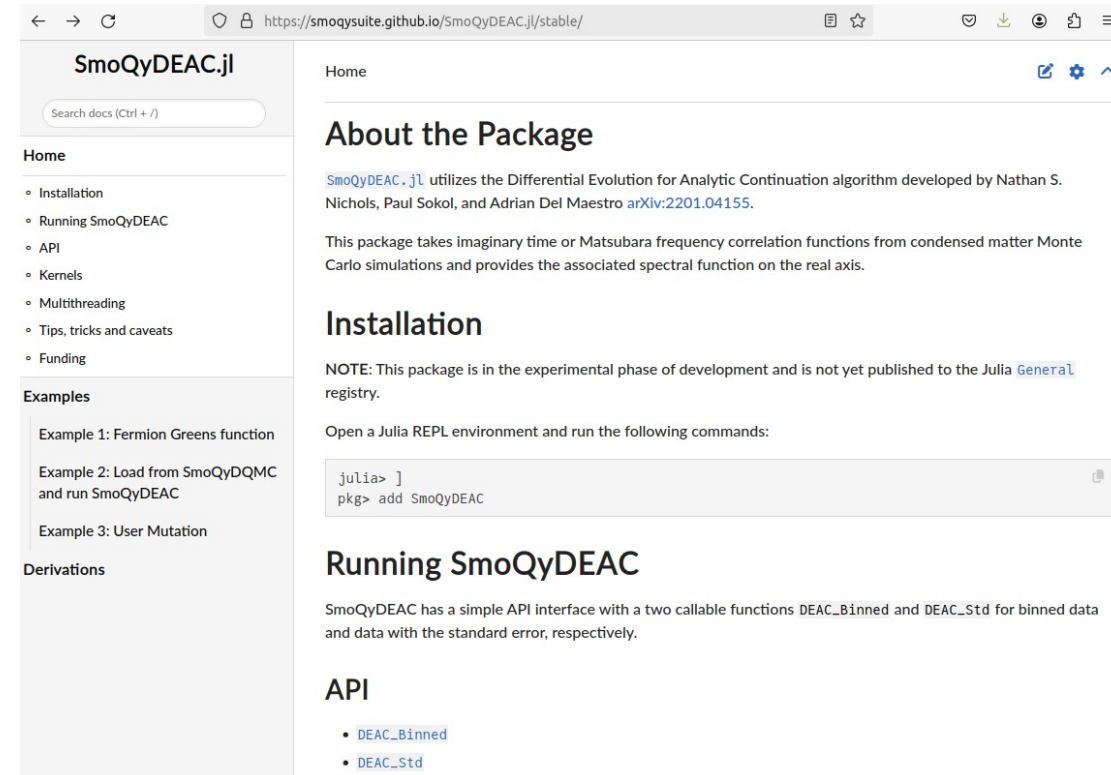
N Nichols et al. Phys. Rev. E 106, 025312 (2022)

DEAC updates



SmoQyDEAC.jl

- Allows Multiple χ^2 targets
- Fitness Floor Finder
- User defined mutations
- Multi-threaded, SIMD accelerated



The screenshot shows the documentation page for SmoQyDEAC.jl, hosted on GitHub. The page has a sidebar on the left with a search bar and a navigation menu. The main content area on the right contains sections for 'About the Package', 'Installation', 'Running SmoQyDEAC', and 'API'.

SmoQyDEAC.jl

Search docs (Ctrl + /)

Home

- Installation
- Running SmoQyDEAC
- API
- Kernels
- Multithreading
- Tips, tricks and caveats
- Funding

Examples

- Example 1: Fermion Greens function
- Example 2: Load from SmoQyDQMC and run SmoQyDEAC
- Example 3: User Mutation

Derivations

About the Package

[SmoQyDEAC.jl](#) utilizes the Differential Evolution for Analytic Continuation algorithm developed by Nathan S. Nichols, Paul Sokol, and Adrian Del Maestro [arXiv:2201.04155](#).

This package takes imaginary time or Matsubara frequency correlation functions from condensed matter Monte Carlo simulations and provides the associated spectral function on the real axis.

Installation

NOTE: This package is in the experimental phase of development and is not yet published to the Julia [General](#) registry.

Open a Julia REPL environment and run the following commands:

```
julia> ]
pkg> add SmoQyDEAC
```

Running SmoQyDEAC

SmoQyDEAC has a simple API interface with a two callable functions `DEAC_Binned` and `DEAC_Std` for binned data and data with the standard error, respectively.

API

- `DEAC_Binned`
- `DEAC_Std`

Using ana_cont and SmoQyDEAC.jl

- `git clone https://github.com/sandimas/UCDavis.git`